Advanced Software Development

Adapted from slides by Ian Sommerville and Edwin Blake

CSC3003S



CSC3003S
Application Architecture Patterns

A. What are Application Architecture Patterns

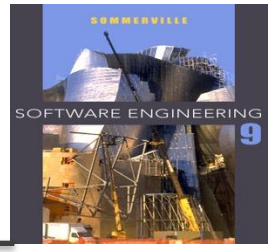
B. Data Processing

C. Transaction Processing

D. Event Processing

E. Language processing

Application systems are intended to meet a business or organizational need

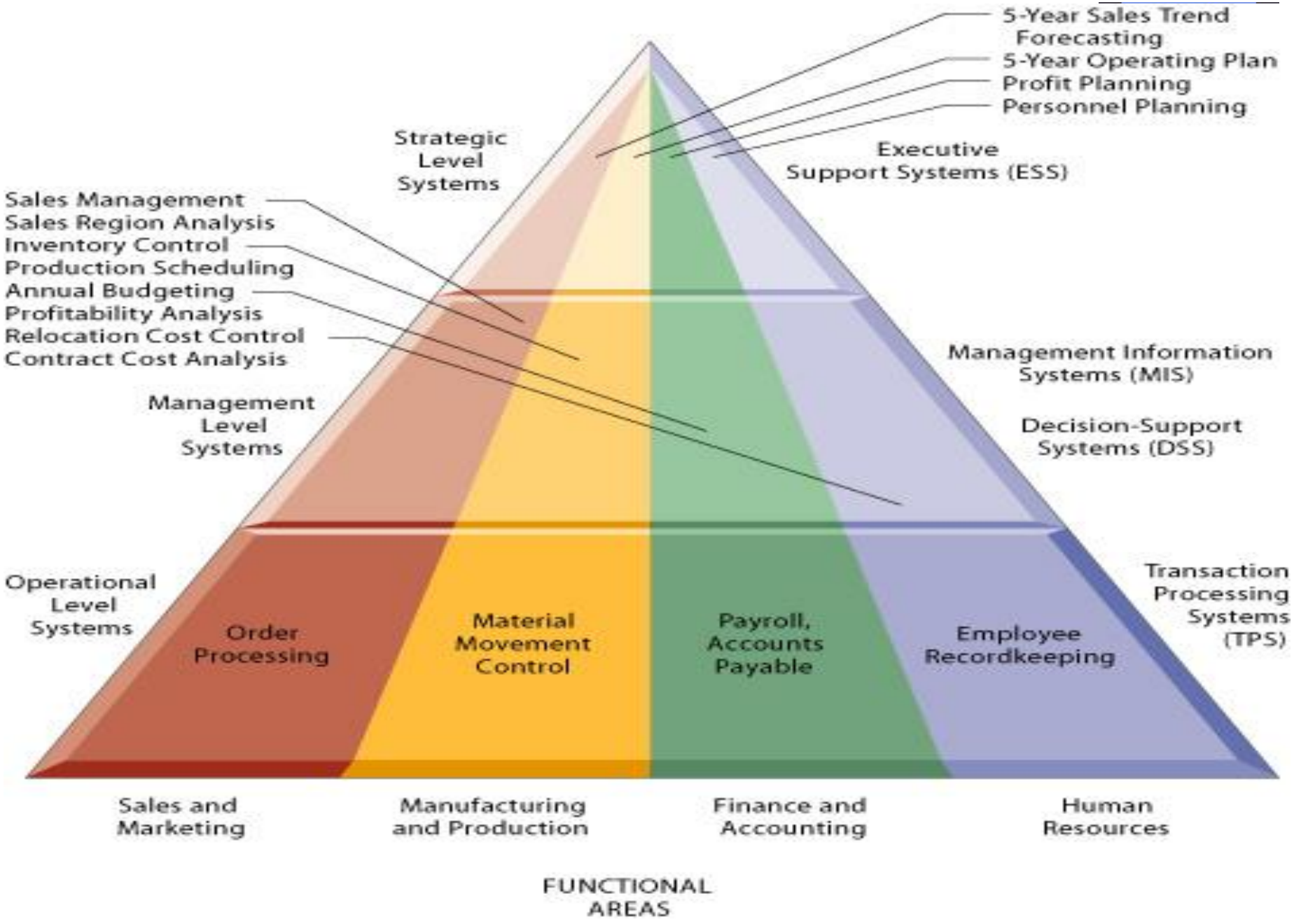




Application architectures encapsulate the principal characteristics of a class of systems.

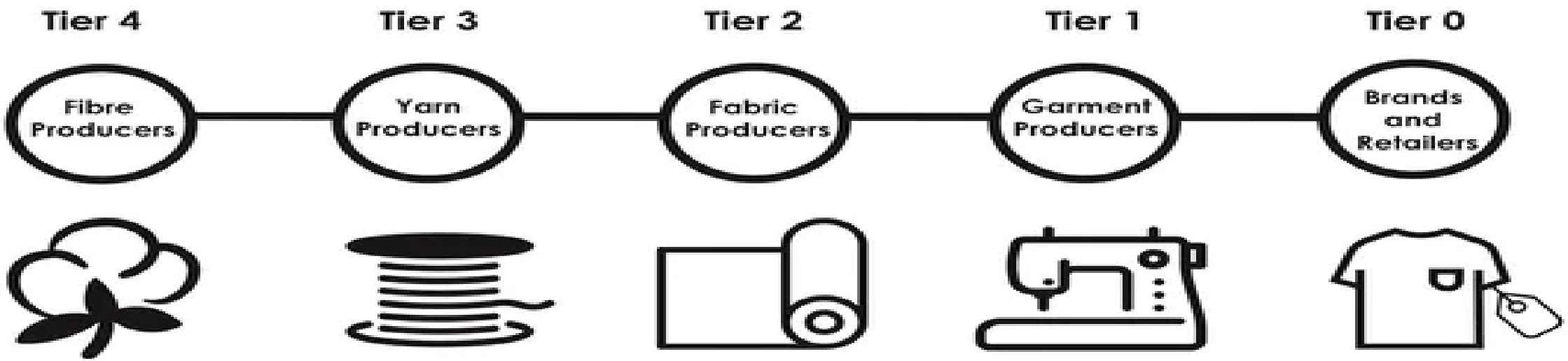


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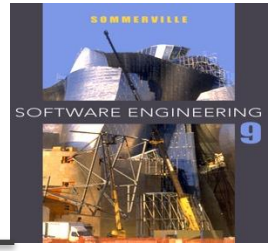


Food Supply Chain





Use of application architectures



- ✧starting
- ✧checking
- ✧organising
- ✧re-using
- ✧discussing



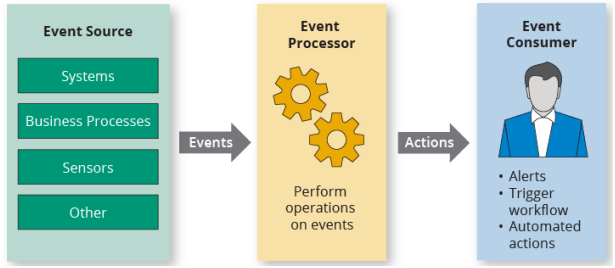
Data processing applications

-Billing systems; Payroll systems



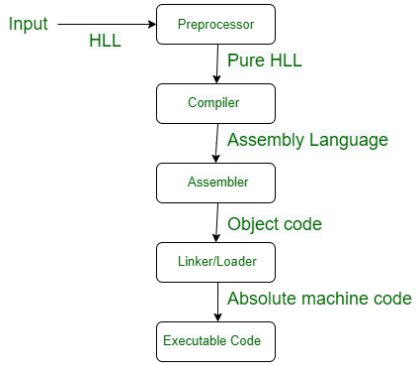
Transaction processing applications

-E-commerce systems; Reservation systems



Event processing systems

-Games; Word processors; Real-time systems



Language processing systems

-Compilers; Command interpreters; ~~English to Zulu …~~



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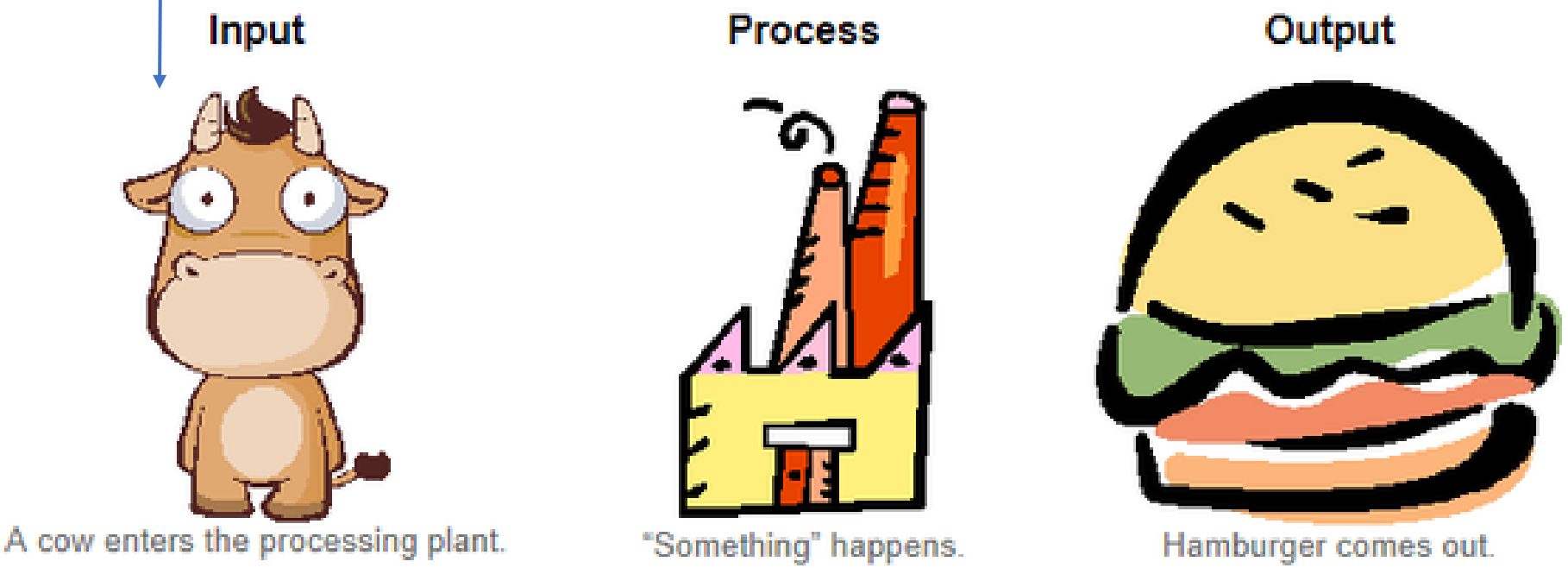
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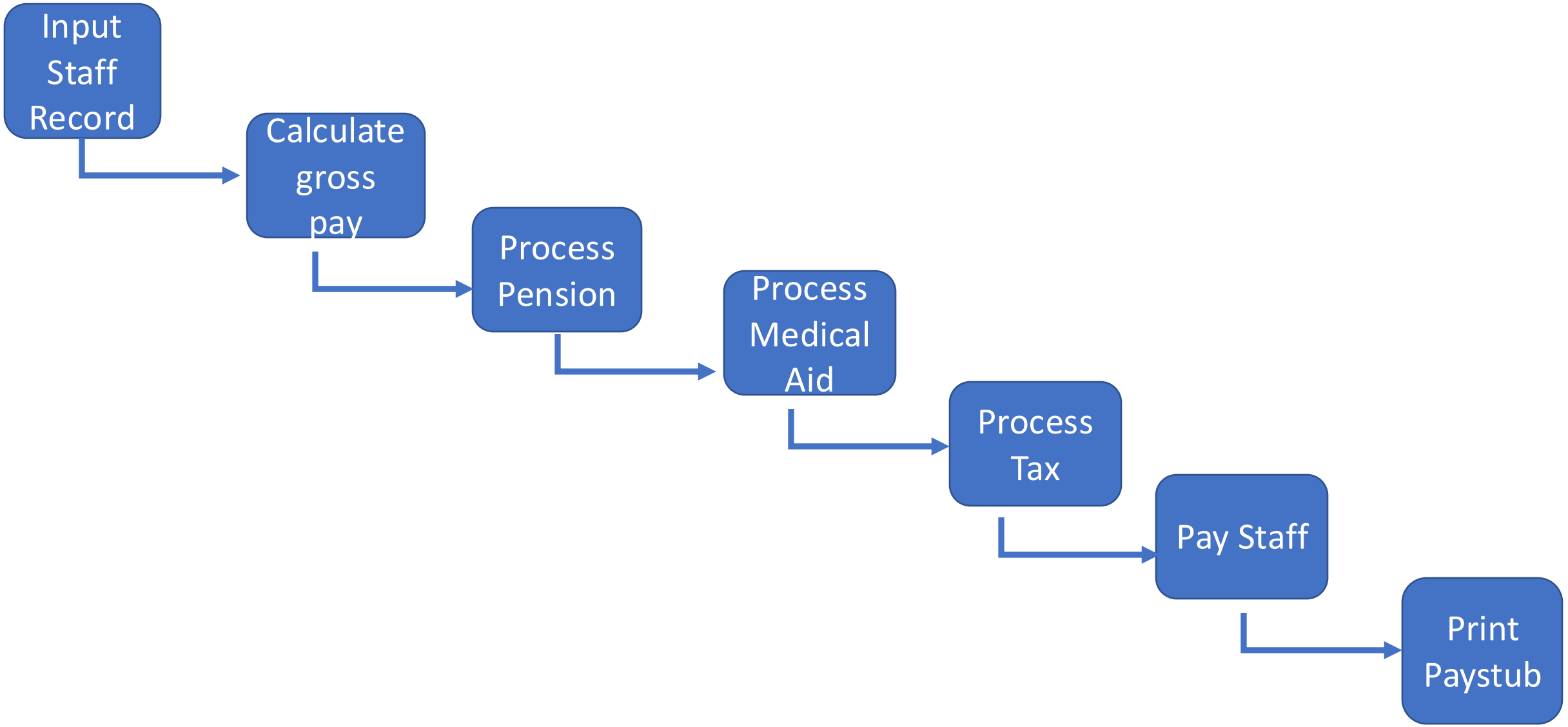
E. Language Processing

input component reads data from a file or database, checks its validity and queues the valid data for processing



output component reads these records, formats them accordingly and writes them to the database or sends them to a printer

process component takes a transaction from the queue (input), performs computations and creates a new record with the results of the computation.


Input Staff Record
Calculate gross pay
Process Pension
Process Medical Aid
Process Tax
Pay Staff
Print Paystub



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Process user requests for information from a database or requests to update the database



Find the times of flights from London to Paris.

Book and pay for flights

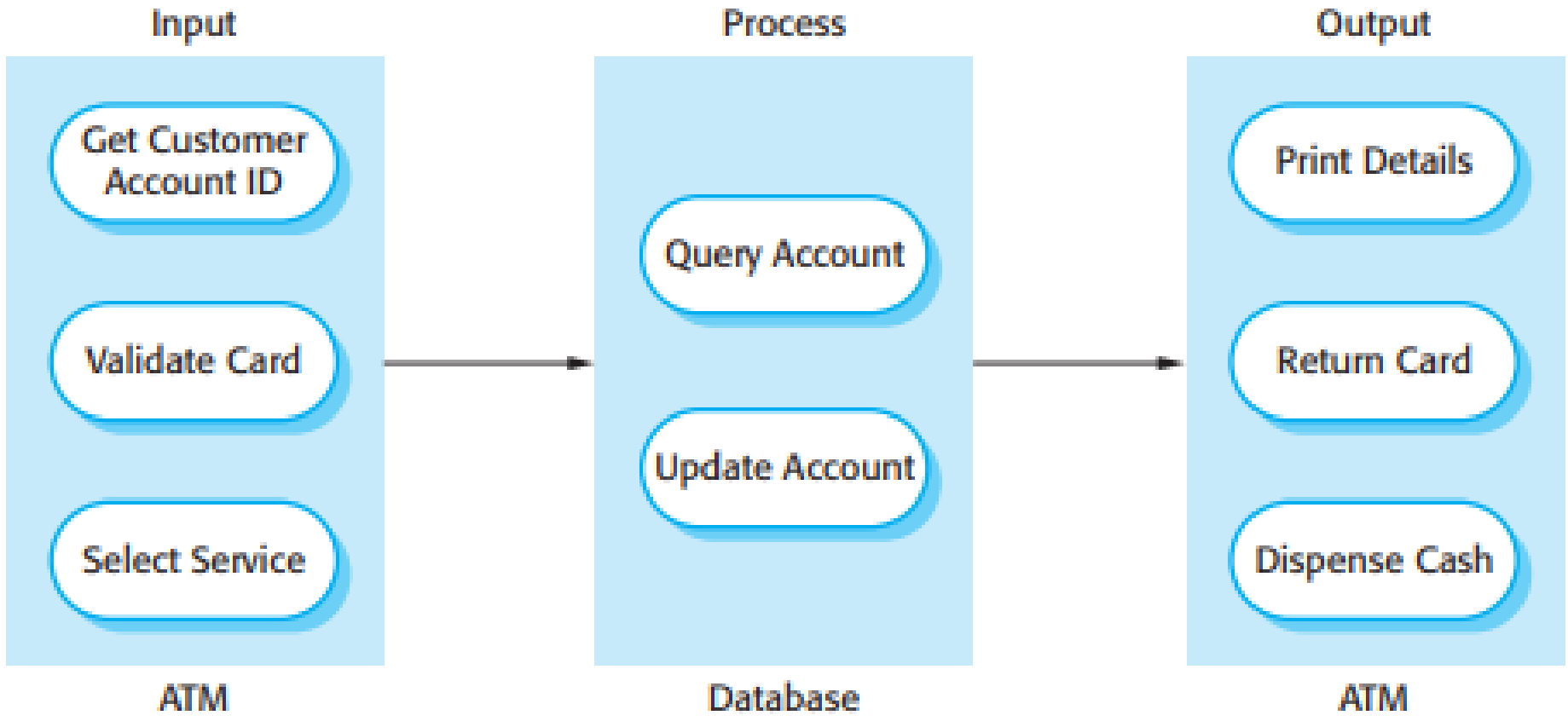
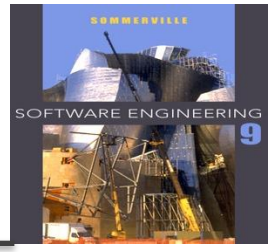
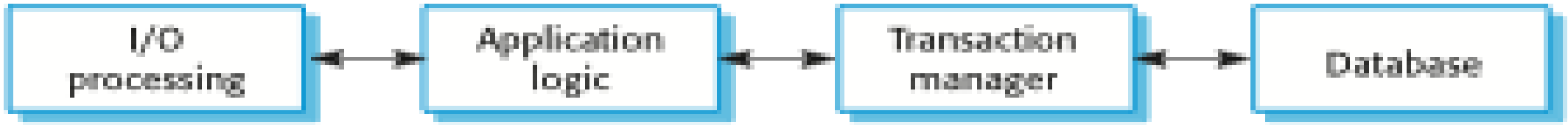


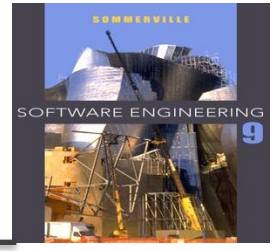
Figure 6.15 The software architecture of an ATM system

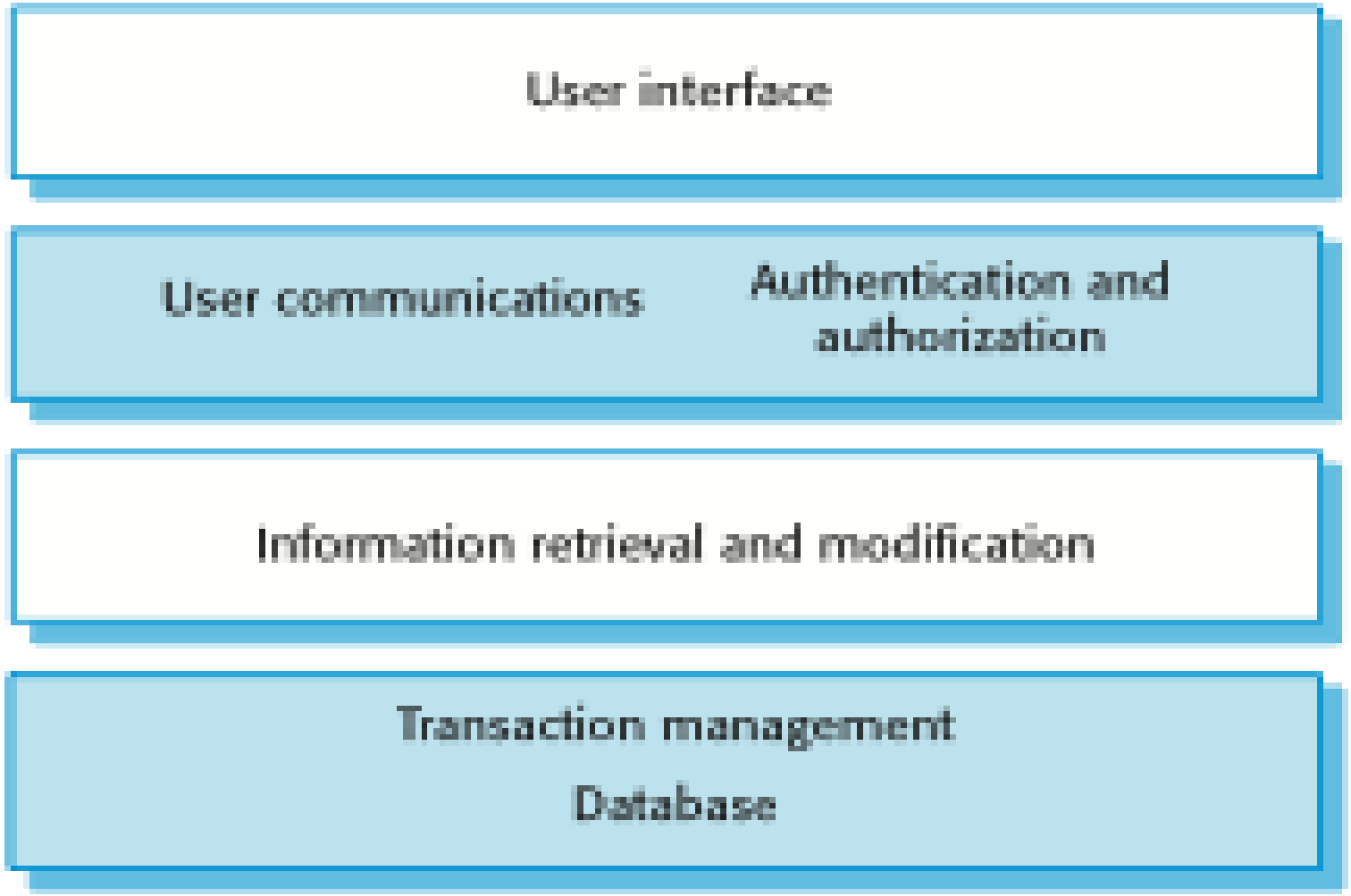
The structure of transaction processing applications



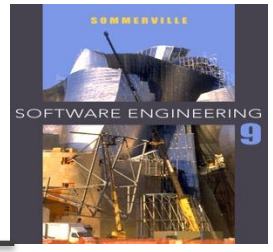


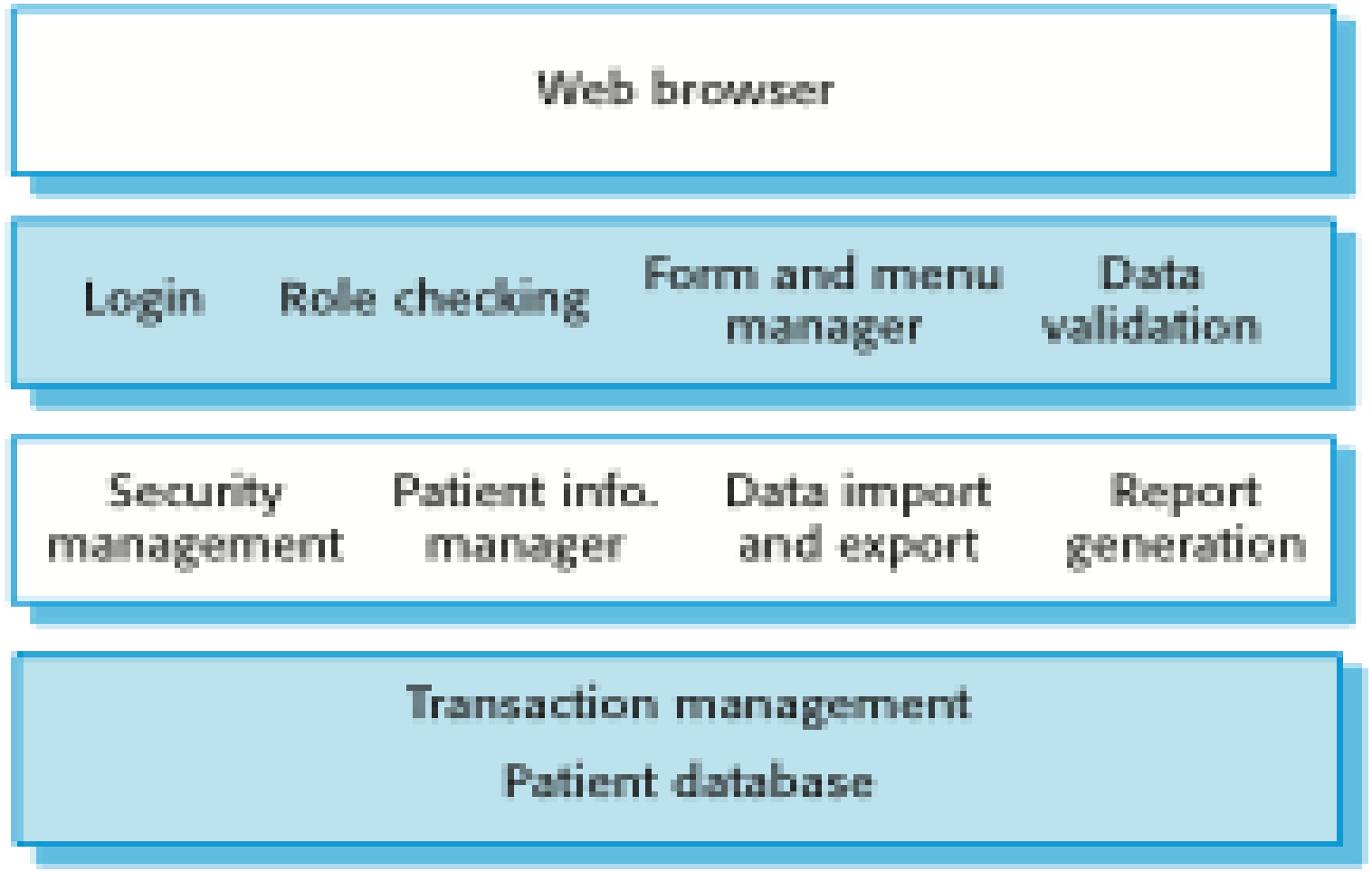
Layered architecture information systems

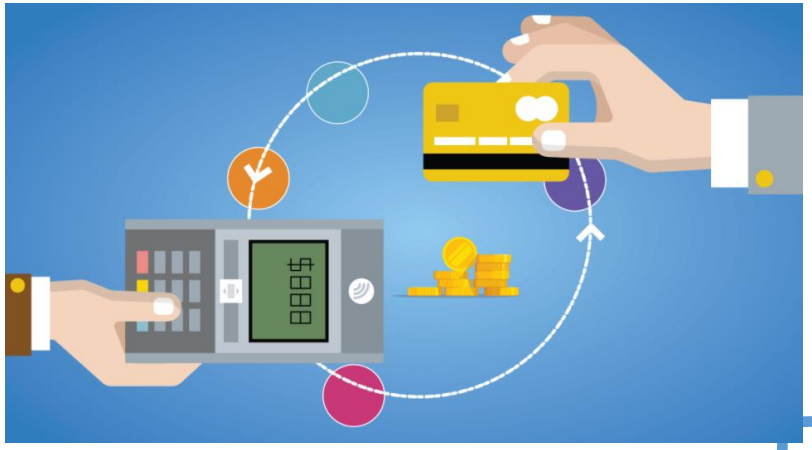




Example: doctors accessing patient records

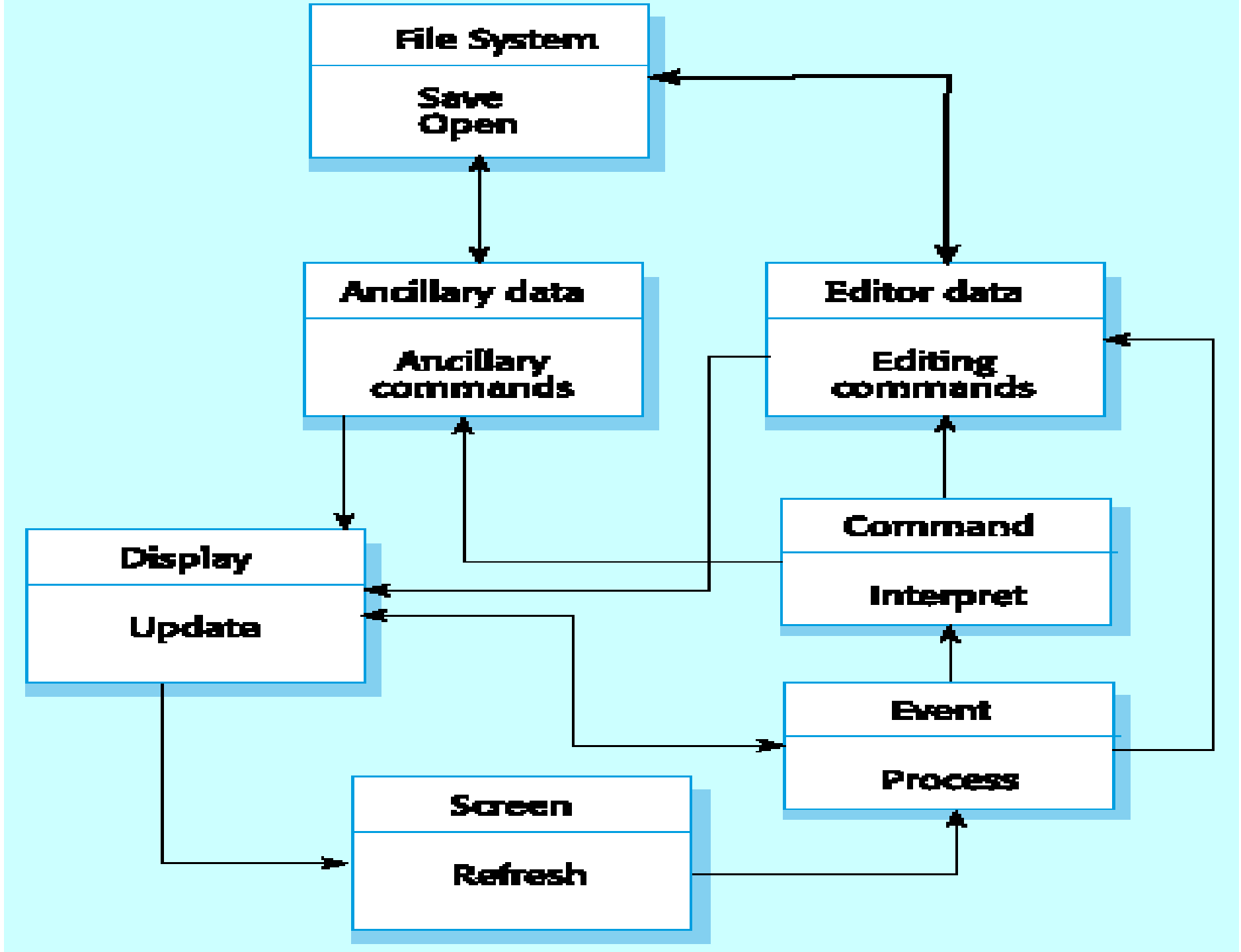






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C. Transaction Processing
D. Event Processing
E. Language Processing

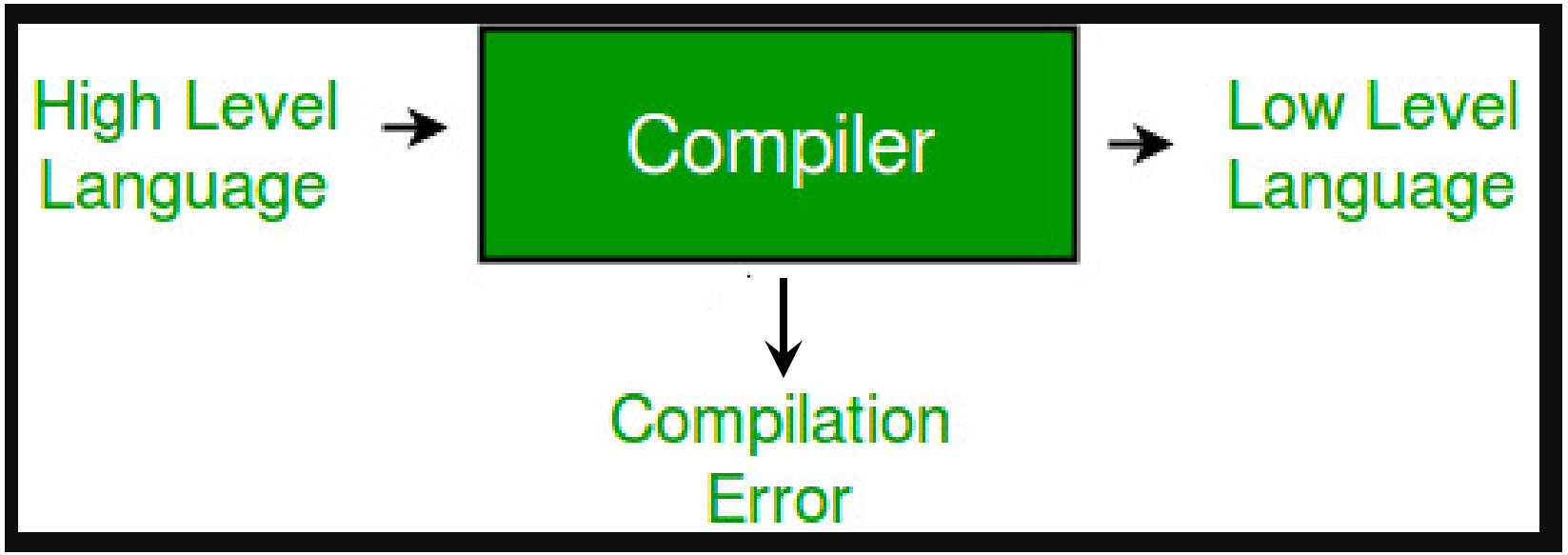

File System
Save
Open
Ancillary data
Ancillary
commands
Editor data
Editing
commands
Display
Update
Command
Interpret
Event
Process
Screen
Refresh


Microsoft
Word

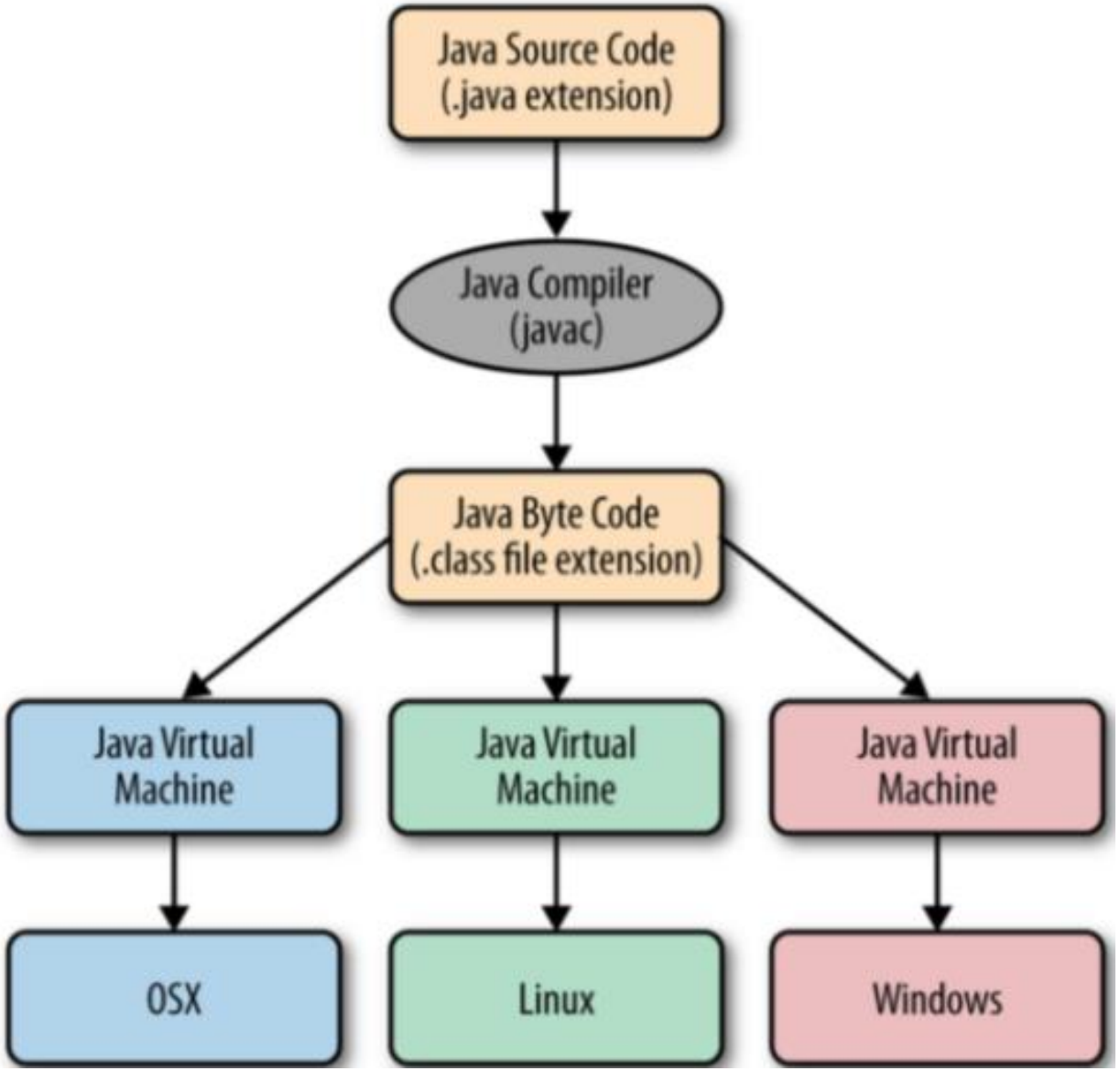


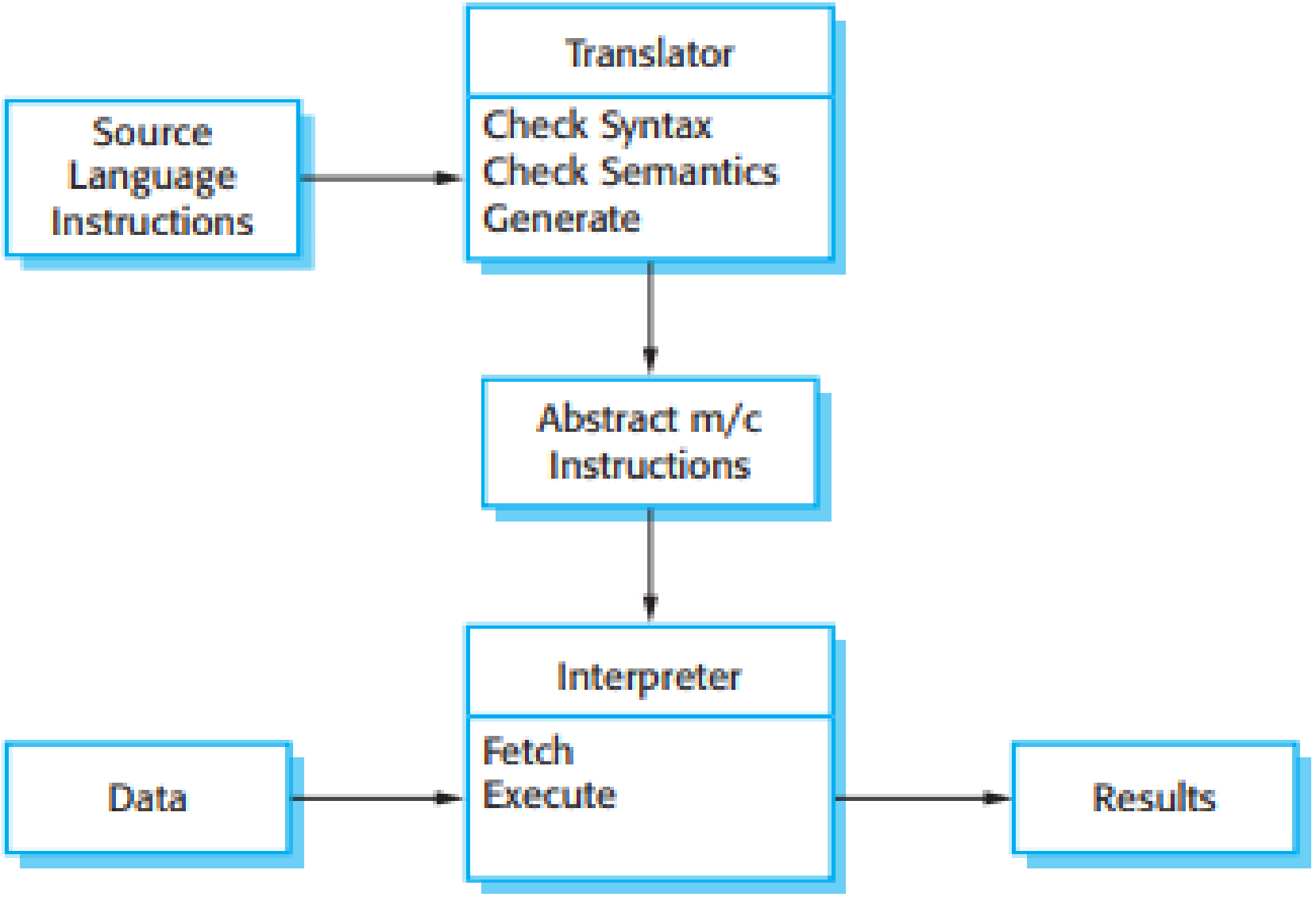
CSC3003S ASD20
Architecture Patterns

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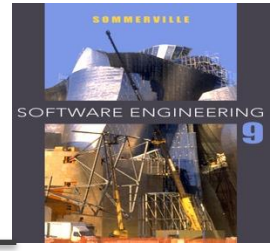

High Level
Language
Compiler
Low Level
Language
Compilation
Error

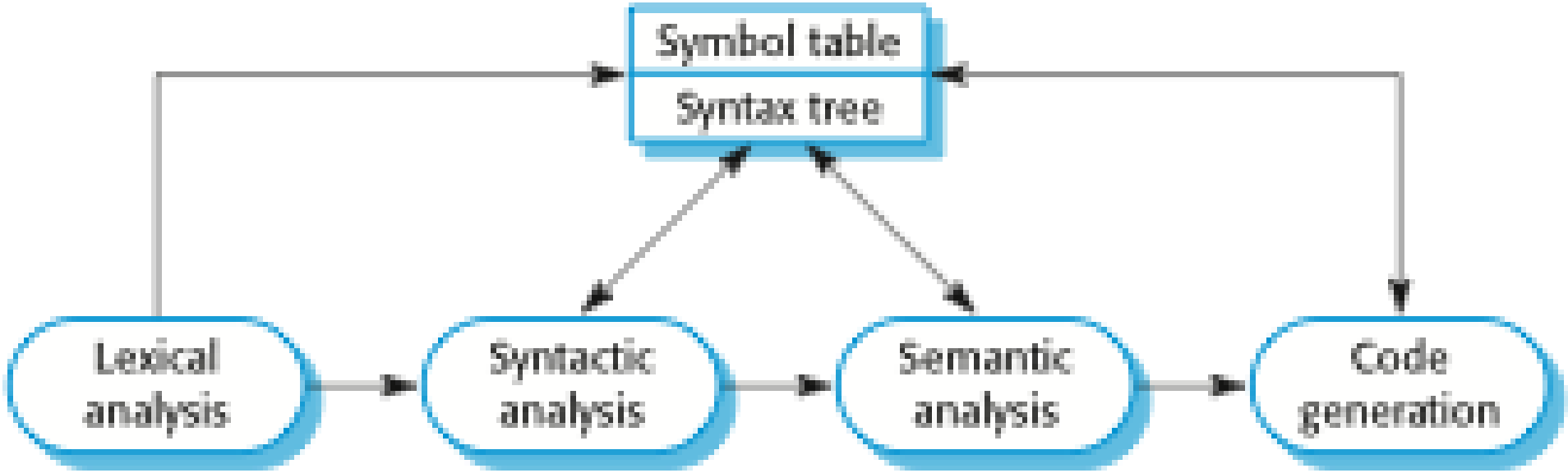
Translate & execute/act



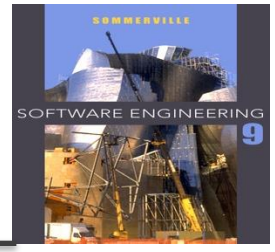

Translator
Source Language Instructions
Check Syntax
Check Semantics
Generate
Abstract m/c Instructions
Interpreter
Data
Fetch
Execute
Results

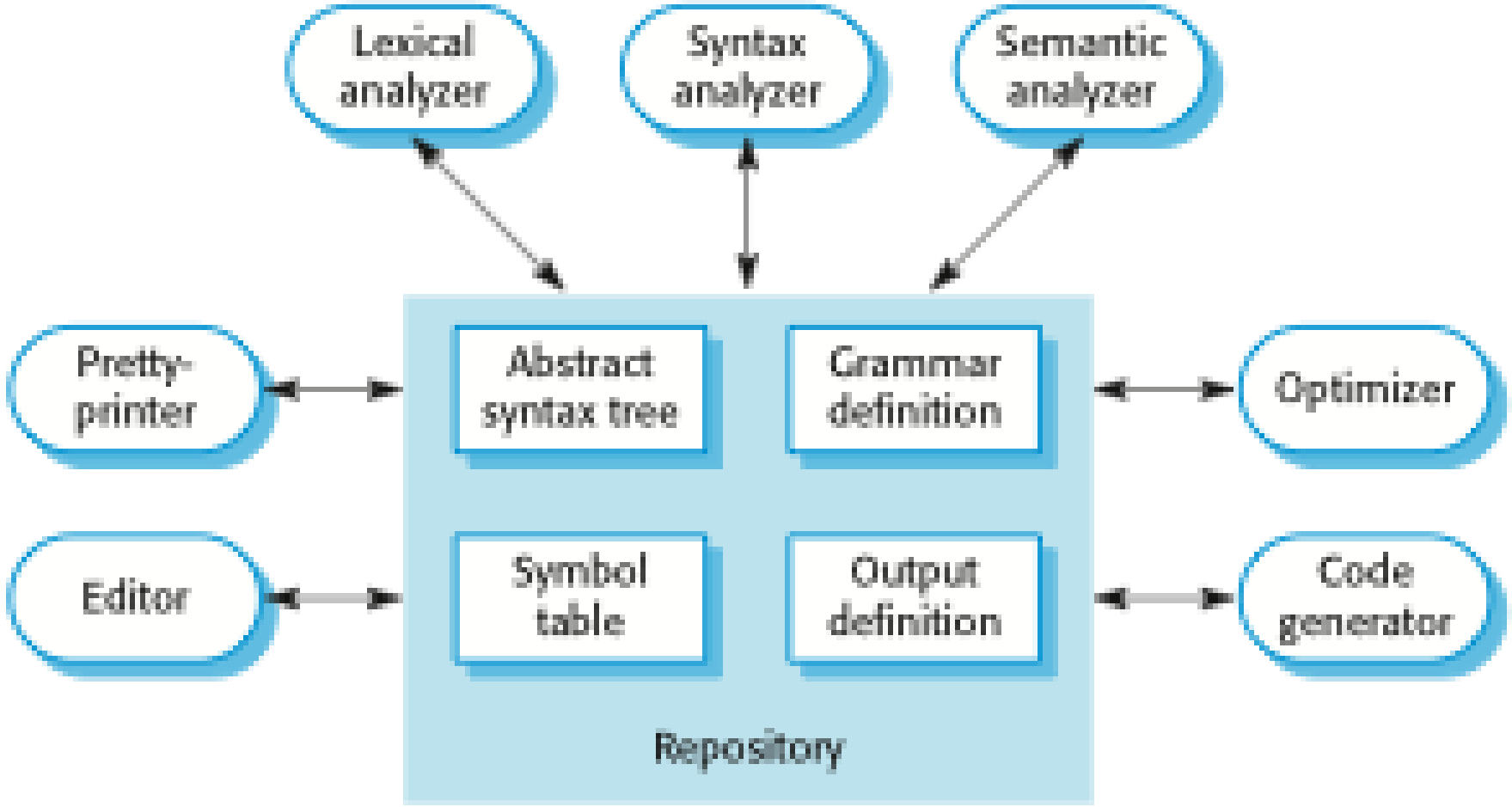
A pipe and filter compiler architecture

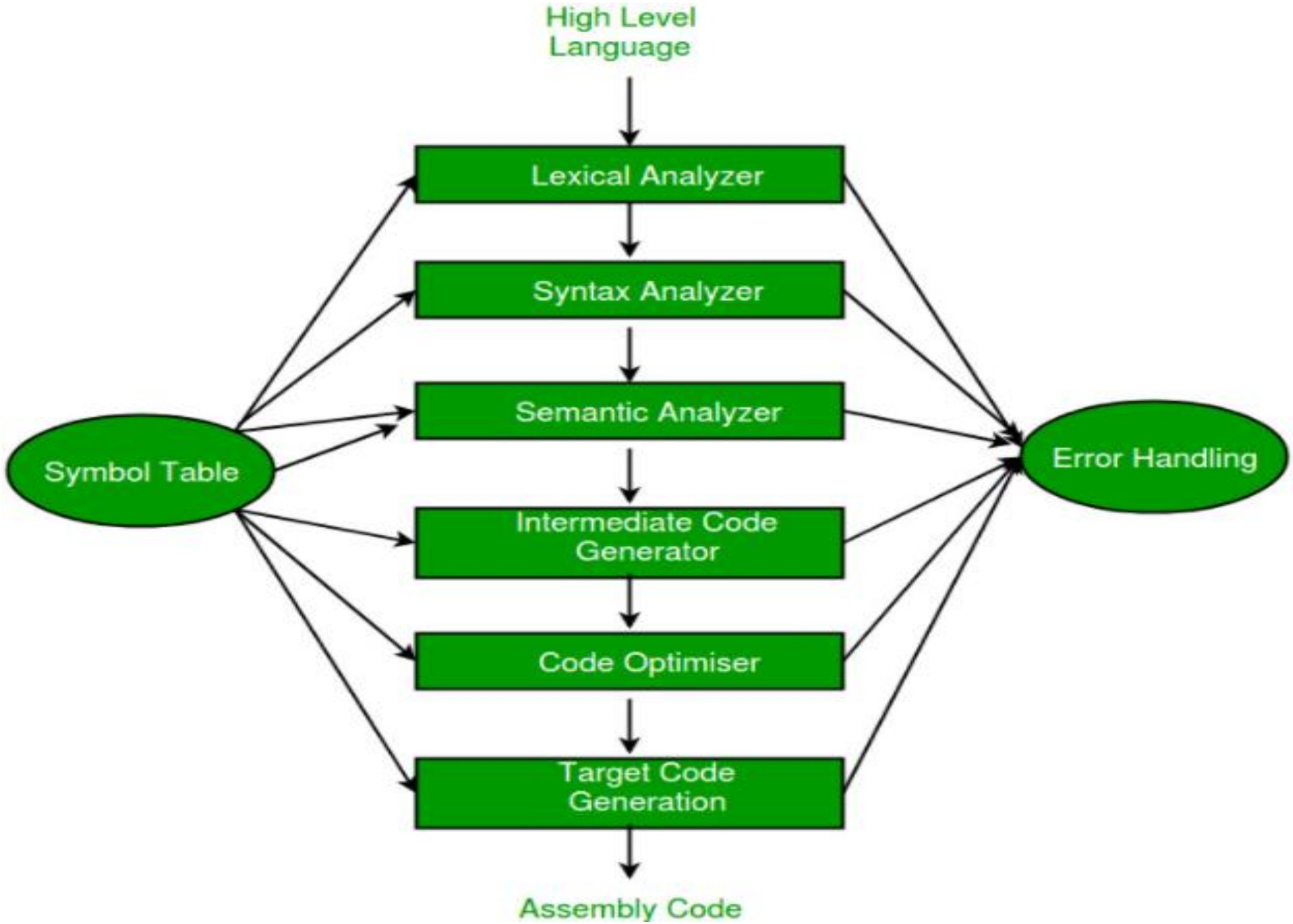


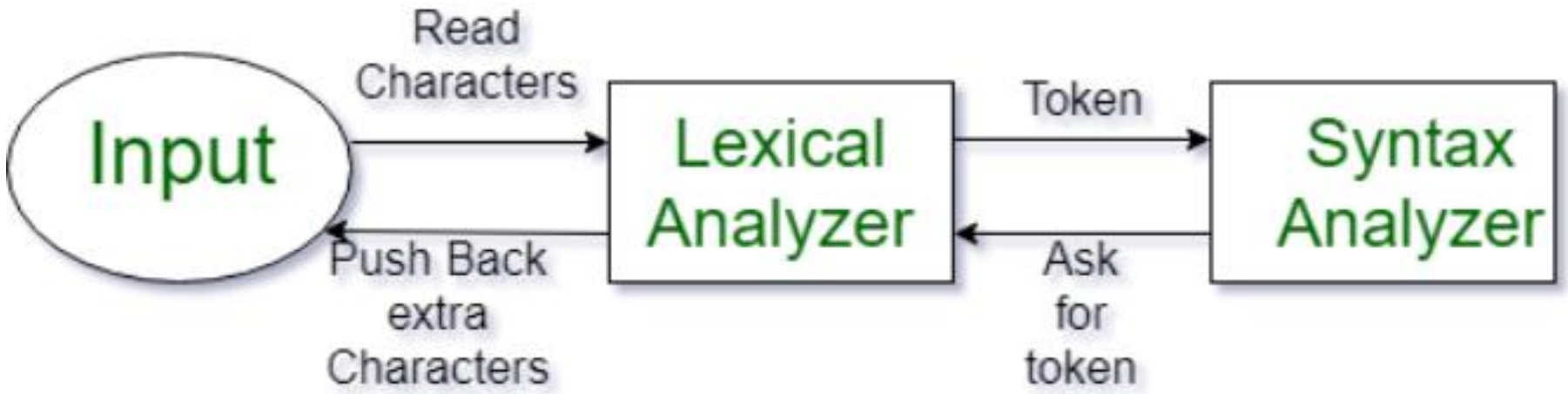


A repository architecture for a language processing system

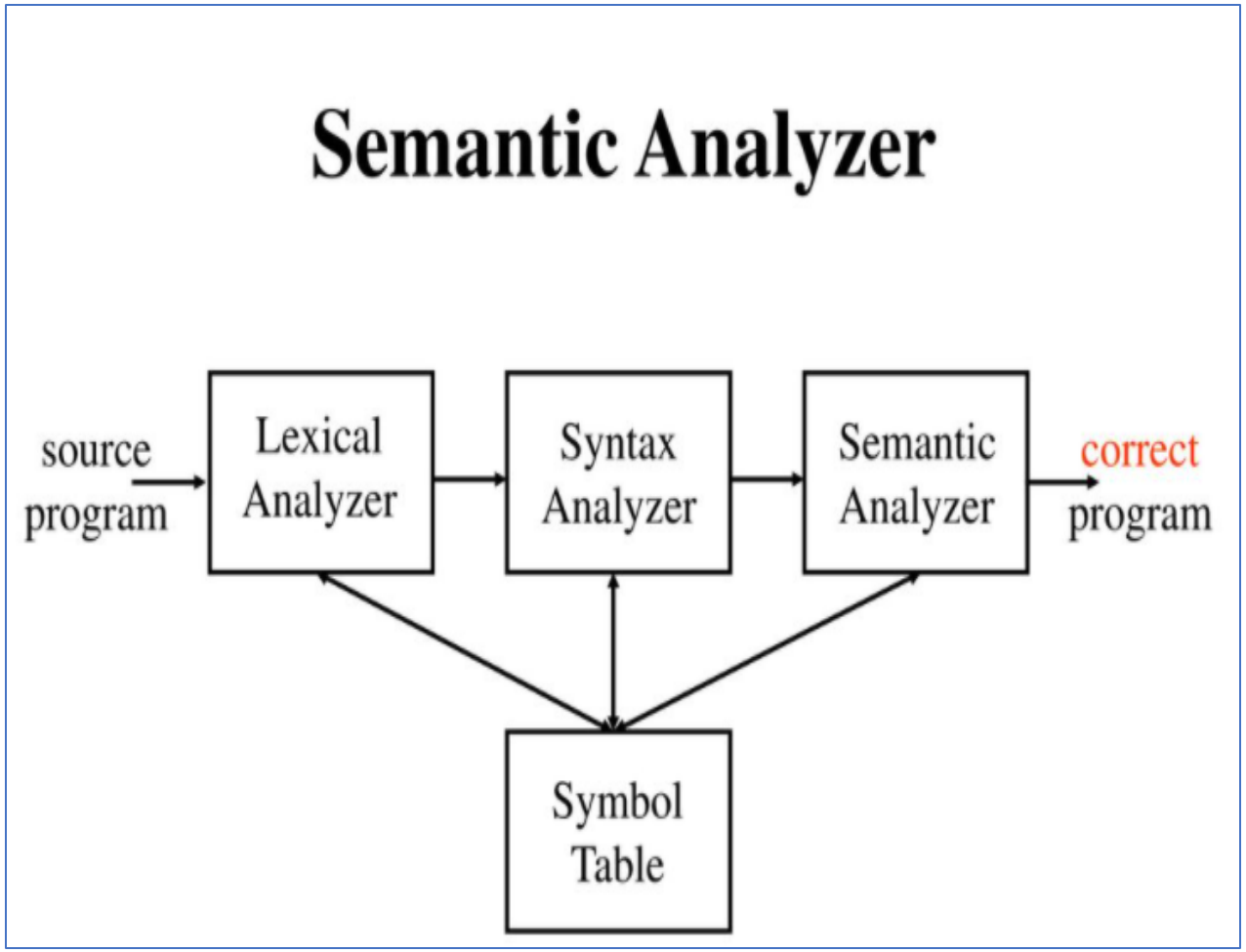


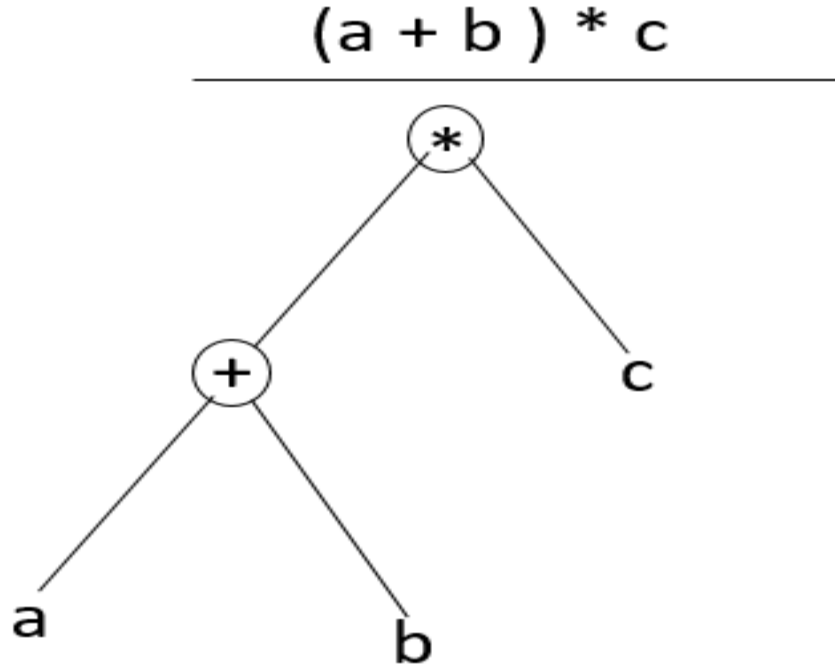



High Level Language
Lexical Analyzer
Syntax Analyzer
Semantic Analyzer
Intermediate Code Generator
Code Optimiser
Target Code Generation
Assembly Code
Symbol Table
Error Handling


Input
Read
Characters
Push Back
extra
Characters
Lexical
Analyzer
Token
Ask
for
token
Syntax
Analyzer


printf ("GeeksQuiz ") ;
1
2
3
4
5


Semantic Analyzer
source program
Lexical Analyzer
Syntax Analyzer
Semantic Analyzer
correct program
Symbol Table


(a + b) * c
*
+
c
a
b

Key points

- Models of application systems architectures help us **understand** and **compare** applications, **validate** application system designs and assess large-scale components for **reuse**.
- **Data processing** system are **batch** systems that allow a database to be accessed and/or modified.
- **Transaction processing** systems are **interactive** systems that allow information in a database to be remotely accessed and modified by a number of users.
- **Language processing** systems are used to translate texts from one language into another. They may include an abstract machine that executes the generated language.